

Rail Crossing Violation Warning – III Cloud-Based Demonstration Final Report

To

Federal Railroad Administration (FRA)

1200 New Jersey Avenue, S.E.
Washington, DC 20590

From

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February 4, 2025

REPORT DOCUMENTATION PAGE				<i>Form Approved OMB No. 07040188</i>	
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1. REPORT DATE (DD-MM-YYYY) 02-04-2025		2. REPORT TYPE Demonstration Report		3. DATES COVERED (From - To) 09-11-2024 – 12-06-2024	
4. TITLE AND SUBTITLE Rail Crossing Violation Warning – III Cloud-Based Demonstration Final Report				5a. CONTRACT NUMBER 693JJ6-22-C-000037	
				5b. GRANT NUMBER	
				5c. PROGRAM ELEMENT NUMBER	
6. AUTHOR(S) Alejandro Sanchez-Badillo 0000-0003-0454-0488 , Gregory Baumgardner 0000-0002-2211-9876 ,				5d. PROJECT NUMBER	
				5e. TASK NUMBER	
				5f. WORK UNIT NUMBER	
7. PERFORMING ORGANIZATION NAME(S) AND ADDRESS(ES) Battelle Memorial Institute 505 King Ave Columbus, OH 43201				8. PERFORMING ORGANIZATION REPORT NUMBER	
9. SPONSORING/MONITORING AGENCY NAME(S) AND ADDRESS(ES) U.S. Department of Transportation Federal Railroad Administration Office of Railroad Policy and Development Office of Research, Development and Technology Washington, DC 20590				10. SPONSOR/MONITOR'S ACRONYM(S)	
				11. SPONSOR/MONITOR'S REPORT NUMBER(S)	
12. DISTRIBUTION/AVAILABILITY STATEMENT					
13. SUPPLEMENTARY NOTES COR: Jared Withers					
14. ABSTRACT The document reports the details and results of the RCVW Cloud-Based System demonstration in the city of Jacksonville, Florida. From September 11, 2024, to December 6, 2024.					
15. SUBJECT TERMS Rail Crossing Violation Warning, RCVW, connected vehicles, CV, Connected Automated Vehicles, CAV, Highway-Rail Intersection, Dedicated Short Range Communications, signal phase and timing, SPaT, Map Data, MAP, Institute of Electrical and Electronics Engineers, IEEE,C-V2X. Cloud Messaging					
16. SECURITY CLASSIFICATION OF:			17. LIMITATION OF ABSTRACT	18. NUMBER OF PAGES 41	19a. NAME OF RESPONSIBLE PERSON
a. REPORT	b. ABSTRACT	c. THIS PAGE			19b. TELEPHONE NUMBER (Include area code)

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Introduction

This document provides the Demonstration report for the deployment of four Rail Crossing Violation Warning (RCVW) Cloud-Based Subsystem (CBS) systems in the city of Jacksonville, FL. This demonstration event took place from 09/11/2024 to 12/06/2024. This report describes the site selection, equipment interface and architecture, data collected, analysis, results and issues encountered during the demonstration period.

Document Organization

This document consists of the following chapters and content.

Referenced Documents identifies the external documentation referenced within this document.

Demonstration Scope provides an overview and general description of the demonstration event

Equipment interface provides a description of the selected sites where the equipment was installed, the hardware architecture and interface diagrams.

Demonstration Tools describes the different tools and methodologies used to gather data as well as to provide a visual representation of the CBS capabilities.

Data gathering analysis and results provides a description of how the data was collected and analyzed.

Demonstration Challenges

Conclusion presents a summary of the results of the demonstration event and potential future opportunities.

Abbreviations and Acronyms.

Referenced Documents

1. Sanchez-Badillo, A., Baumgardner, G (2023). Rail Crossing Violation Warning - Concept of Operations. Federal Railroad Administration,
2. Sanchez-Badillo, A., Baumgardner, G (2023). Rail Crossing Violation Warning. System Requirements Specification. Federal Railroad Administration.
3. Baumgardner, G., Sanchez-Badillo, A. (2020). Rail Crossing Violation Warning - Architecture and Design Specifications.
4. American Association of State and Highway Transportation Officials. (2018). A Policy on Geometric Design of Highways and Streets. AASHTO.

Demonstration Scope

The RCVW was created as a Connected Vehicle (CV) system designed to monitor the status of a railroad preemption signal sent by a Highway-Rail Intersection (HRI). Railroad preemption is used when a signalized intersection is nearby an HRI with the purpose of preventing a train colliding with a vehicle stopped within the crossing. The Manual on Uniform Traffic Control Devices (MUTCD) states that when a signalized intersection is within 200 feet of an HRI, the use of a preemption signal should be considered. When a train approaches the HRI, the HRI controller preempts the traffic signal controller thereby initiating a sequence of steps to clear any potential highway vehicle traffic stopped within the collision zone of the HRI and preventing more vehicles approaching the crossing. Similar to a traffic signal controller, the RCVW generates and broadcast changes within Signal, Phase, and Timing (SPaT) messages whenever there is a change in the railroad preemption status.

The RCVW now includes a CBS which receives and reports these changes in the railroad preemption signal (amongst other things) to authorized subscribers over the internet. A Roadside-based subsystem detects the changes in the preemption signal indicating an activation of the monitored HRI's warning devices. This information is sent out to the CBS which then receives it, stores it, and pushes it or makes it available (depending on the subscriber type) to its authorized subscribed users.

The main goal of the CBS is to allow the broadcast of timely information about an activation of an HRI warning device to any user subscribed to receive related notifications. This capability would allow drivers whom the HRI is part of their route, to choose alternative ways or postpone the start of a trip minimizing delays and congestion. In the case of emergency vehicles, being aware of the status of the HRI aids emergency responders to plan their approach and avoid delays. A freight dispatcher would be able to monitor routes to avoid delays and optimize schedules.

For this demonstration, four HRI locations in the city of Jacksonville, FL were equipped with the RCVW CBS with the goal of achieving the following:

- Alert cloud subscribers of the activation/deactivation of the monitored HRI protective devices
- Inform cloud subscribers of the communication health between the RCVW unit monitoring the HRI and the CBS
- Provide cloud subscribers the following static HRI related information
 - Crossing Identification Number (ID)
 - City
 - County
 - Street
 - Railroad Organization
 - Emergency Contact Number
 - Railroad Contact Number
 - Latitude and Longitude
 - Mile Marker
 - Distance to Highway
 - Speed limit
 - Crossing Angle

- Crossing Position
- Gate Presence
- Flashing Light Presence

This information was provided in the form of a Graphical User Interface (GUI) intended to simulate a Traffic Management Center (TMC). A TMC is a central hub from which traffic is monitored and managed in real-time. The primary purpose of a TMC is to optimize the flow of traffic, improve safety, and minimize the effects of incidents that occur on the road.

Key functions of a TMC include:

1. Monitoring traffic conditions: TMCs use various technologies like Closed-Circuit Television (CCTV) cameras, traffic detectors, and other data sources to monitor traffic flow and road conditions.
2. Incident management: In case of accidents, breakdowns, or other incidents, TMCs coordinate the response, which may involve dispatching emergency services, providing real-time information to drivers, and rerouting traffic.
3. Traffic signal control: TMCs often have the ability to remotely control traffic signals to optimize traffic flow, especially during peak hours or special events.
4. Information dissemination: TMCs provide real-time traffic information to the public through various channels like variable message signs, websites, mobile apps, and radio broadcasts.
5. Planning and strategy: TMCs also play a role in long-term traffic planning and strategy, using the data they collect to identify trends and potential improvements.

By having access to timely HRI-related information a TMC would be capable of:

1. Safety Monitoring: Monitoring train movements at HRIs allows the TMC to ensure safety for both vehicles and pedestrians. By detecting when the HRI active warning devices are activated, they can provide timely warnings to drivers and pedestrians, preventing accidents.
2. Traffic Flow Optimization: Knowing when HRI warning devices are activated helps on the management of traffic flow. The TMC can adjust signal timings, reroute vehicles, or provide real-time information to drivers, minimizing congestion and delays.
3. Emergency Response Coordination: In case of emergencies (e.g., stalled trains, derailments, or accidents), the center needs accurate information about train locations. This enables efficient coordination with emergency services and ensures a swift response.
4. Infrastructure Maintenance: Monitoring HRI warning device activation helps track wear and tear on the infrastructure. Regular maintenance can be scheduled based on usage patterns, reducing downtime and ensuring safe crossings.

The GUI used for the demonstration presented a map view of the city of Jacksonville, FL. The map indicated the location of HRIs in the area as well as the four locations equipped with the RCVW CBS.

For this demonstration, an authorized user of the TMC would click on one of the sites accessing the HRI related information. When the CBS receives data from the RCVW roadside based subsystem (RBS) indicating that the HRI warning devices are active, the GUI issues an audible alert and changes the icon of the location to indicate the such. When the warning devices deactivate, the icon returns to its original shape and color.

When a monitored HRI site loses connectivity with the CBS, the GUI changes the icon of the location, indicating a problem is detected. The user can then click on the site and information related to the situation is presented.

All the data related to the HRI (activations, communication loss, deactivations, etc.) is stored in the CBS for future analysis and retrieval. See the Demonstration Tools section for more details on the GUI.

The demonstration was geared to highlight the performance of the CBS by alerting subscribers to the activation of the HRI warning devices and to collect data to perform a latency analysis of the system.

Participation

The following entities were essential in the deployment of this demonstration

- The Federal Railroad Administration is the agency funding the development of the RCVW CBS as well as the implementation of the demonstration.
- Florida Department of Transportation (FDOT) District Two provided coordination and selection of the HRI sites as well as the installation and retrieval of data.
- Battelle Memorial Institute Software and Systems engineers were behind the development of the RCVW CBS.
- Trimble provided the Battelle team with a developer's license and an Application Programming Interface (API) key that enabled the use of Trimble maps to create the GUI tool.

Planning

The Battelle team reached out to FDOT District Two with the proposal to deploy the CBS at four HRI locations that matched the following characteristics:

- HRIs that experiences 10 or more activations in a period of 24 hours.
- The selected HRI is located within 200 feet of a signalized intersection.
- The traffic signal controller at the signalized intersection is interconnected with railroad signal preemption from the HRI controller.
- The Intersection experiences frequent traffic disruptions due to HRI warning device activations.

Four locations in the city of Jacksonville, FL. were identified:

Table 1. HRI Locations in Jacksonville. FL

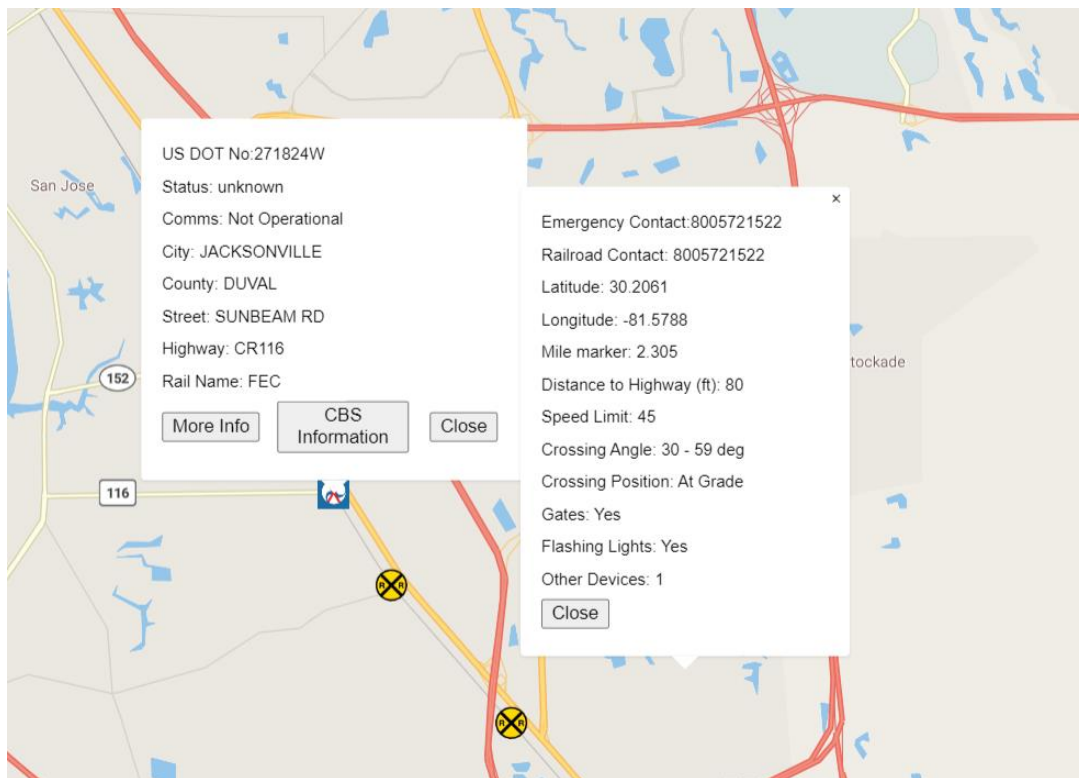
Crossing ID	City	Street	Latitude /Longitude
621223F	Jacksonville,FL	San Juan Avenue	30.282196, -81.720559
621221S	Jacksonville,FL	St. Johns Ave	30.2885320, -81.7206070
271824W	Jacksonville,FL	Sunbeam Rd	30.2060770, -81.5787700
621275X	Jacksonville,FL	S Edgewood Ave	30.3269015, -81.7308415

According to FDOT, CSX freight and Amtrack passenger trains travel over these HRIs. Based on crossing data provided by FDOT there is an average of ten train activations of the HRI warning devices in these locations as a result of an approaching train every day.



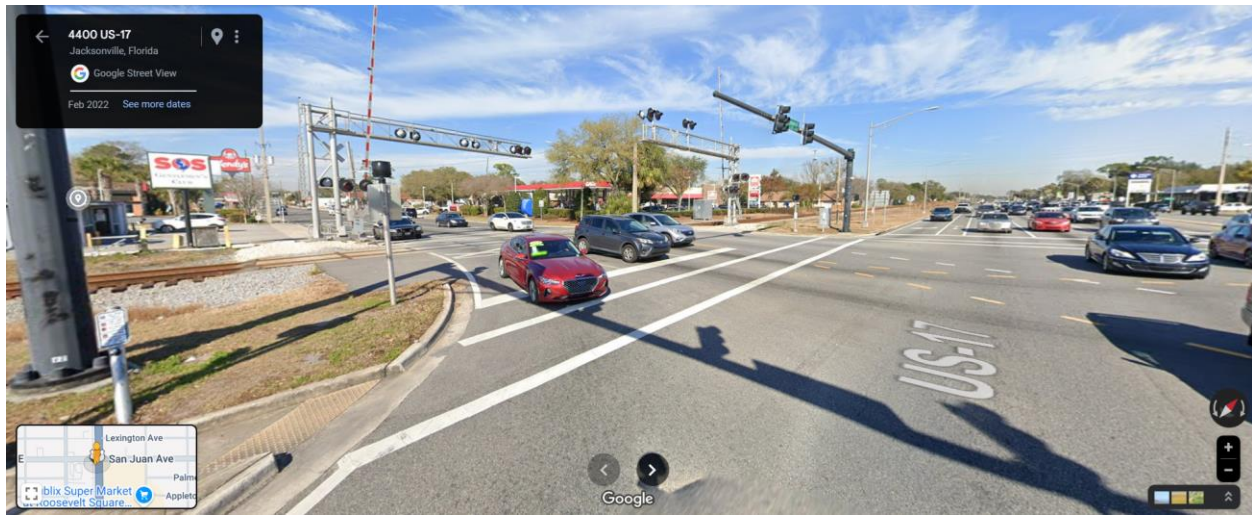
Source: Google Maps

Figure 1. Sunbeam Road HRI Street View



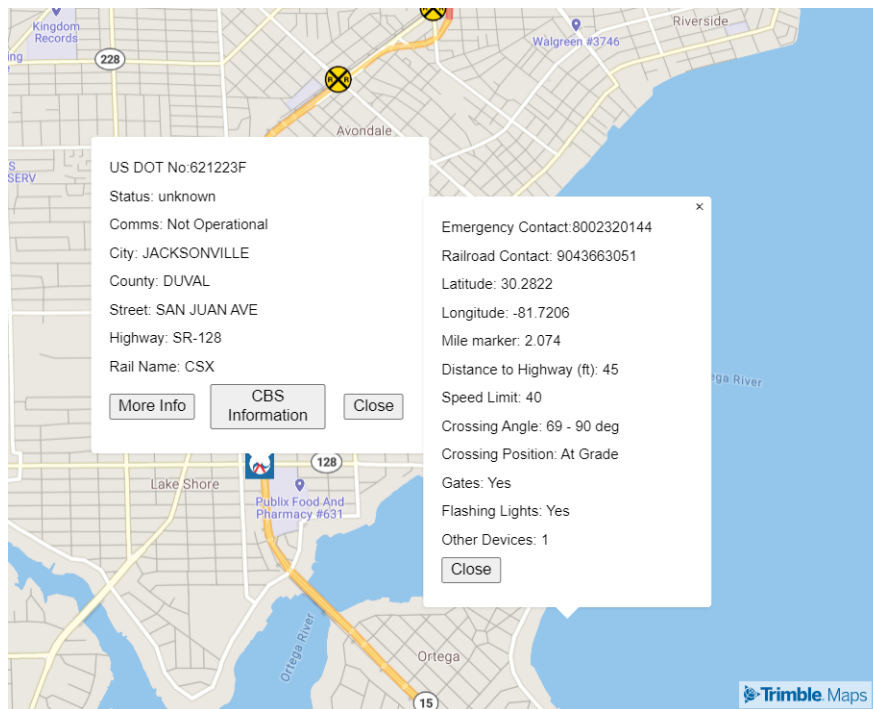
Source: Battelle

Figure 2. Sunbeam Road TMC Application View



Source: Google Maps

Figure 3. San Juan Avenue HRI Street View



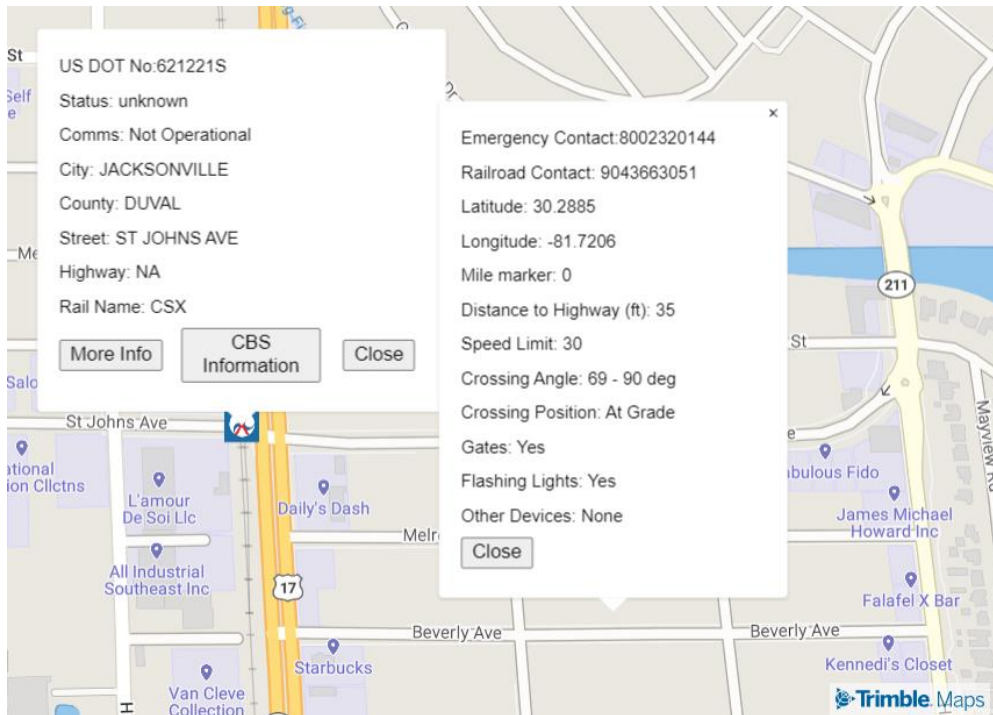
Source: Battelle

Figure 4. San Juan Avenue TMC Application View



Source: Google Maps

Figure 5. St. Johns Avenue HRI Street View



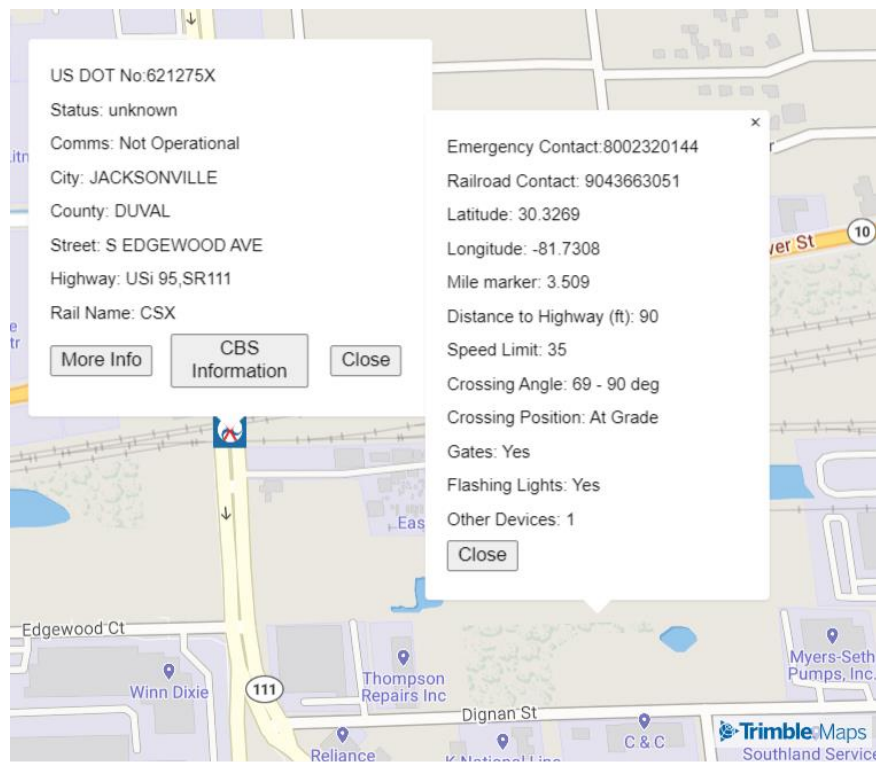
Source: Battelle

Figure 6. St. Johns Avenue TMC Application View



Source: Google Maps

Figure 7. Edgewood Avenue HRI Street View



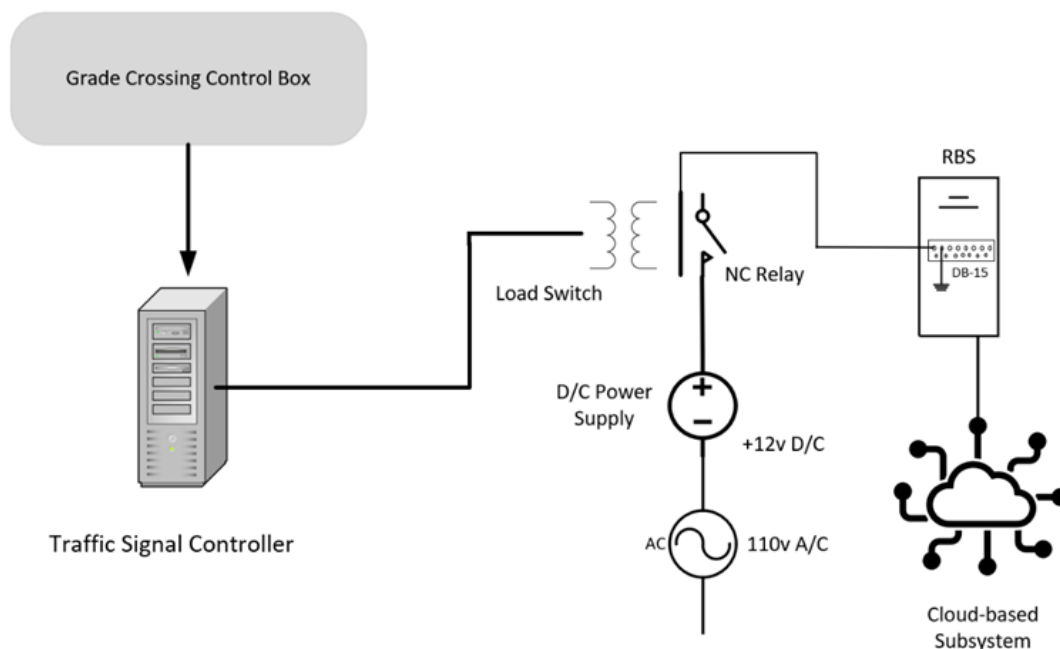
Source: Battelle

Figure 8. Edgewood Ave TMC Application View

Equipment Interface

The RCVW RBS is designed to receive preemption signals directly from the HRI controller to detect warning device activation. The RBS follows the common track circuit design where a constant 24 Volts Direct Current (VDC) signal indicates warning devices inactive while a drop in the voltage indicates warning devices active. This signal is received by the RBS via a DB-15 digital I/O (DIO) interface.

For this demonstration, the Battelle team selected sites where an HRI controller is interconnected with a Traffic Signal Controller via a preemption signal. It was planned to connect the RCVW RBS to the Traffic Signal Controller preemption signal input. However, due to concerns from the city of Jacksonville of the RCVW interacting with the traffic signal phases, it was decided to change the interface architecture and instead receive a signal from a load switch within the Traffic Controller cabinet (see Signal Preemption Interface for more details). When the preemption signal from the HRI controller is removed due to the arrival of a train, the Traffic Signal Controller reads this and sends a signal to a load switch within the controller cabinet. The load switch then closes and completes a 120 VAC circuit which activates the corresponding traffic light phases dedicated to stopping traffic towards the HRI (Red light to this approach) and clearing traffic coming out of it (Green light to traffic to clear HRI). For this demonstration, the Battelle team used this 120 Volts Alternate Current (VAC) signal to open a normally closed relay that provides the required 12 VDC signal which simulates the railroad preemption signal coming into the RCVW RBS (via a power supply). When this happens, the load switch opens the 12 VDC circuit to the RBS, thus generating a change in the SPaT message indicating the activation of the HRI warning devices. (Figure 9)



Source: Battelle

Figure 9. RBS Interface with Traffic Signal Controller

Due to concerns from FDOT

The RBS at the four sites used a dedicated wired internet connection to the CBS provided by FDOT. However, at the St. Johns location, there was the need to use a portable wireless router over cellular network (MiFi) instead. See the Data Analysis Data Analysis section for details. The CBS had each of these locations pre-programmed in the cloud with its corresponding HRI ID, SAE J2735 HRI Geometric

Message (MAP) file, Latitude and Longitude ready to receive HRI related information. These sites were pre-loaded as well with HRI related attributes from the FRA database. See Table 2, Table 3, Table 4, Table 5 and Table 6 for details.

Table 2. Known HRI ID – Message Data

Data Elements	Data Type
HRI ID`	Int: 32 bit
Latitude of fix	Float: nnnn.nn
N or S	Char: A
Longitude of fix	Float: nnnnn.nn
E or W	Char: A

Table 3. HRI Crossing Header - Data Elements

Data Elements	Data Type
HRI ID	Int: 32 bit
HRI Protective warning device Present	Char: A
Agency ID	Int: 32 bit
County Code	Char: NNNNN
Crossing ID	Char: NNNNNNN
Published Report Base ID	Int: 32 bit
State Cde	Char:NN
Created	Char: ddmmyy hhmmss
Created By	Char: 50 byte limit
Last Updated	Char: ddmmyy hhmmss
Last Updated By	Char: 50 byte limit

Table 4. HRI Highway Traffic Control Device - Data Elements

Data Elements	Data Type
HRI ID	Int: 32 bit
AdvWarn	Char: 32 byte limit
AwdlDate	Char: ddmmyy hhmmss
AwHornChk	Char:A
AwHornlDate	Char: ddmmyy hhmmss
Bells	Int: 32 bit
Bkl_FlashPost	Char: A
CFlashType	Char: 32 byte limit
Channel	Char: 32 byte limit
EnsSign	Char: A
Exempt	Char: A
FlashNov	Int: 32 bit
FlashOther	Int: 32 bit
FlashOtherDes	Char: 256 limit
FlashOv	Int: 32 bit
FlashPai	Int: 32 bit
FlashPost	Int: 32 bit
FlashPostType	Char: 32 byte limit
GateConf	Char: 32 byte limit
GateConfType	Char: 32 byte limit
GatePed	Int: 32 bit
Gates	Int: 32 bit
HwtrfPsig	Char: A
HwtrfPsigIndis	Int: 32 bit
HwtrfPsigdis	Int: 32 bit
HwynrSig	Char: A
HwyTrafSignl	Char: A
Intrprmp	Char: 32 byte limit

Data Elements	Data Type
Led	Char: 256 byte limit
Low_Grnd	Char: A
Low_GrndSigns	Int: 32 bit
MonitorDev	Char: 32 byte limit
NoSigns	Char: A
OthDes1	Char: 10 byte limit
OthDes2	Char: 10 byte limit
OthDes3	Char: 10 byte limit
OthSign	Char: A
OthSign1	Int: 32 bit
OthSign2	Int: 32 bit
OthSign3	Int: 32 bit
PaveMrkIDs	Char: 32 byte limit
PrempType	Char: 32 byte limit
PrvxSign	Char: A
ReportBaseld	Int: 32 bit
Sdl_FlashPost	Char: A
SpecPro	Char: 20 byte limit
StopStd	Int: 32 bit
XBuck	Int: 32 bit
YieldStd	Int: 32 bit
WdCode	Char: N
Created	Char: ddmmyy hhmmss
Last Updated	Char: ddmmyy hhmmss
Last Updated By	Char: 50 byte limit

Table 5. HRI Location and Classification - Data Elements

Data Elements	Data Type
HRI ID	Int: 32 bit
BlockNumb	Char: 6 byte limit
CityCD	Char: 10 byte limit
CntyCD	Char: 10 byte limit
DevelTypId	Char: 32 byte limit
Highway	Char: 256 byte limit
HscoRrid	Char: NNNN
HwyCont	Char: 10 byte limit
Latitude	Char: 256 byte limit
LLsource	Char: 10 byte limit
Longitude	Char: 256 byte limit
MultFrmsFiled	Char: A
Nearest	Char: 6 byte limit
OpenPub	Char: A
PolCont	Char: 10 byte limit
PosXing	Char: 32 byte limit
Railroad	Char: 32 byte limit
ReportBaseld	Int: 32 bit
RrCont	Char: 10 byte limit
RrID	Char: 256 byte limit
RrMain	Char: 32 byte limit
RrNarr	Char: Max
RrNarr1	Char: 256 byte limit
RrNarr2	Char: 256 byte limit
RrNarr3	Char: 256 byte limit
RrNarr4	Char: 256 byte limit
SameInd	Char: A
SameRr1	Char: 32 byte limit

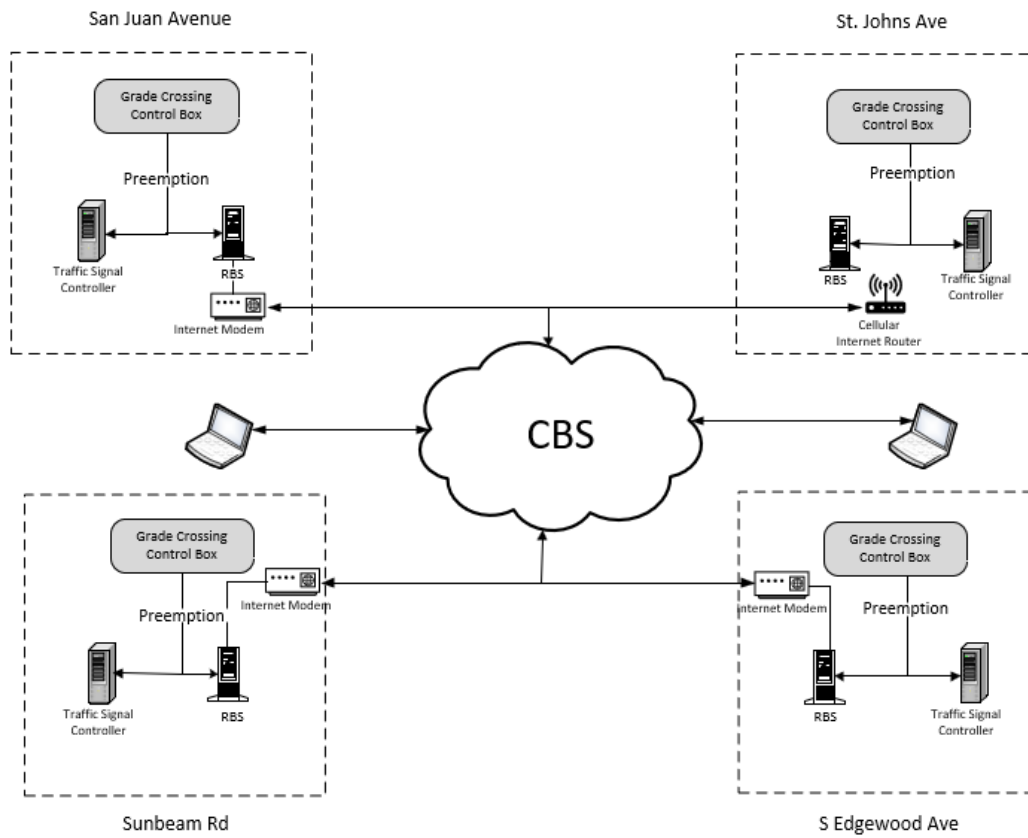
Data Elements	Data Type
SameRr2	Char: 32 byte limit
SameRr3	Char: 32 byte limit
SameRr4	Char: 32 byte limit
SepInd	Char: A
SepRr1	Char: 32 byte limit
SepRr2	Char: 32 byte limit
SepRr3	Char: 32 byte limit
SepRr4	Char: 32 byte limit
SfxHscoRrid	Char: 4 byte limit
StateCD	Char: NN
StNarr	Char: Max
StNarr1	Char: 256 byte limit
StNarr2	Char: 256 byte limit
StNarr3	Char: 256 byte limit
StNarr4	Char: 256 byte limit
Street	Char: 256 byte limit
Ttstn	Char: 6 byte limit
TtstnNam	Char: 256 byte limit
TypeTrnSrvclId	Char: 32 byte limit
TypeXing	Char: 32 byte limit
Whistban	Char: 32 byte limit
WhistDate	Char: ddmmyy hhmmss
XingAdj	Char: A
XingOwnr	Char: 32 byte limit
XngAdjNo	Char: 7 byte limit
XPurpose	Char: 32 byte limit
Created	Char: ddmmyy hhmmss
Created By	Char: 50 byte limit
Last Updated	Char: ddmmyy hhmmss

Data Elements	Data Type
Last Updated By	Char: 50 byte limit

Table 6. HRI Physical Characteristics - Data Elements

Data Elements	Data Type
HRI ID	Int: 32 bit
ComPower	Char: A
Downst	Char: A
HwynDist	Int: 32 bit
HwyNear	Char: A
HwyPved	Char: A
Illumina	Char: A
ReportBaseId	Int: 32 bit
TrafficLn	Int: 32 bit
TraflnType	Char: 32 byte limit
XAngle	Char: 32 byte limit
XSurfaceIDs	Char: 32 byte limit
XSurfDate	Char: MMYYYY
XSurfLength	Int: 32 bit
XSurfWidth	Int: 32 bit
XSurfOthr	Char: 256 byte limit
Created	Char: ddmmyy hhmmss
Created By	Char: 50 byte limit
Last Updated	Char: ddmmyy hhmmss
Last Updated By	Char: 50 byte limit

Figure 10 shows a graphic representation of the RCVW configuration for this demonstration.



Source: Battelle

Figure 10, Jacksonville, FL RCVW configuration

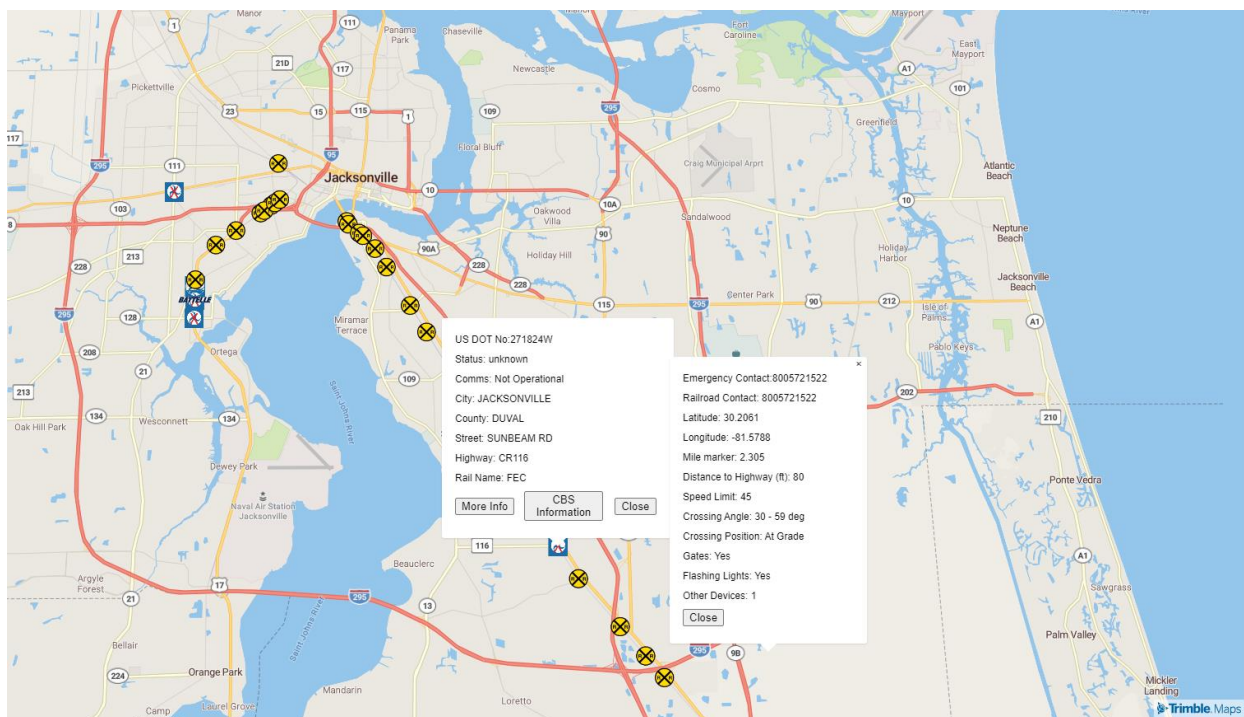
Demonstration Tools

The GUI used to display the location and status of the HRIs was a web service created specifically for this demonstration. The Trimble maps development platform was used to display a map of the area, HRI icons and information for each of the locations. This service allows the user to zoom in and out of the location, move and center the map to a particular point and provides a graphic and audio alert when a monitored HRI detected the arrival of a train. The goal for this tool was to serve as a simulated TMC monitoring HRIs on real time. This TMC acted as an authorized subscriber of the CBS and was monitoring the four equipped RBS. Users with the proper credentials were able to access the service remotely and were presented with a graphical representation of the area where the monitored HRIs are located as well as real-time status of the four equipped locations.

In detail, the GUI enabled the following capabilities:

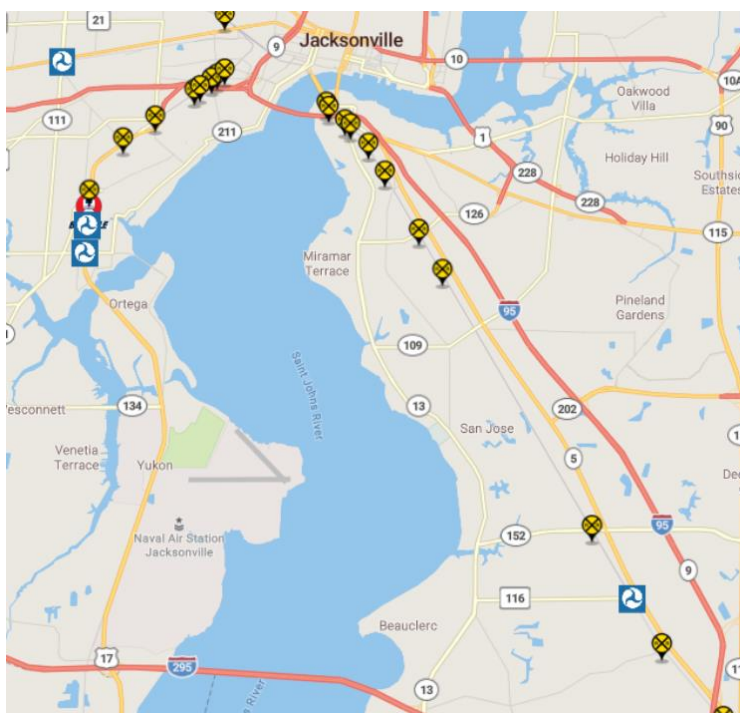
1. Graphic representation of the four RCVW-equipped and several non-RCVW-equipped HRIs
 - All RCVW-equipped HRIs were represented on a map by specific distinguishable icons.
 - All non-RCVW-equipped HRIs were represented on a map by specific distinguishable icons.
2. Graphic representation of the current communications health of the HRI monitored equipment
 - The color of the monitored HRI icons changed depending on the status of the communications link between the CBS and the HRI.
3. Graphic representation of the current activation status of the HRI active warning devices of the monitored HRIs
 - The color and icon of the monitored HRI icons changed depending on the status of the HRI warning devices.
4. Audible alerts
 - When the CBS detected the change in the railroad preemption signal on a monitored HRI, a short audio alert was issued.
5. Information of all HRIs
 - HRI ID
 - HRI Location
 - Activation Status
 - Type of warning devices present
 - Any other information that is presented in the FRA data dictionary can be accessed by the user

In addition, the GUI provided a user interface that allowed a user to simulate a change in the railroad preemption signal or the simulation of a communication problem at a particular HRI. This allowed the user to manually simulate activation and observe the CBS functionality on demand.



Source: Battelle

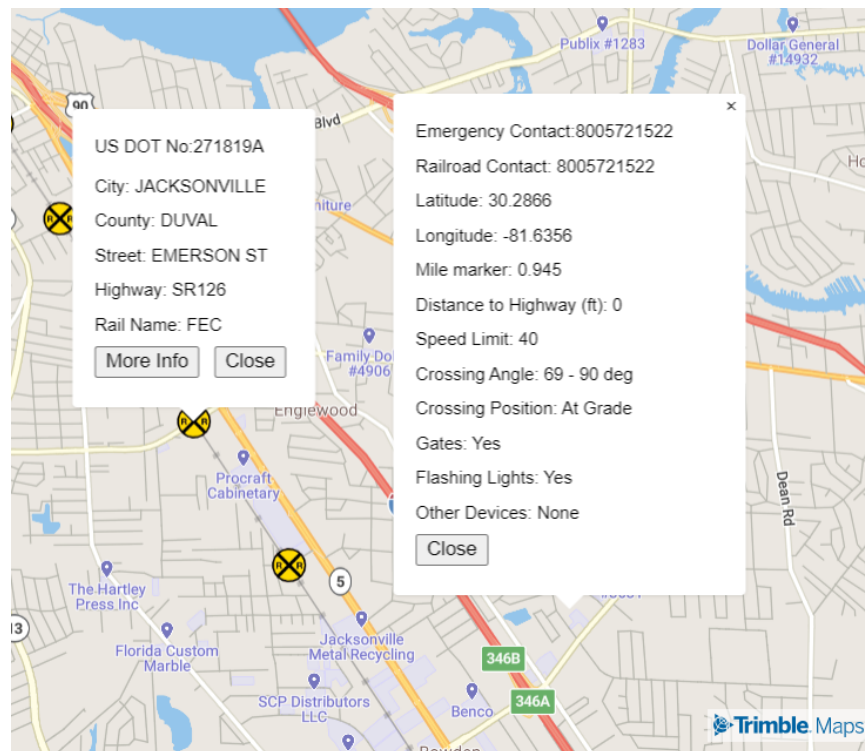
Figure 11. Simulated Traffic Management Center



Source: Battelle

Figure 12. Simulated Traffic Management Center

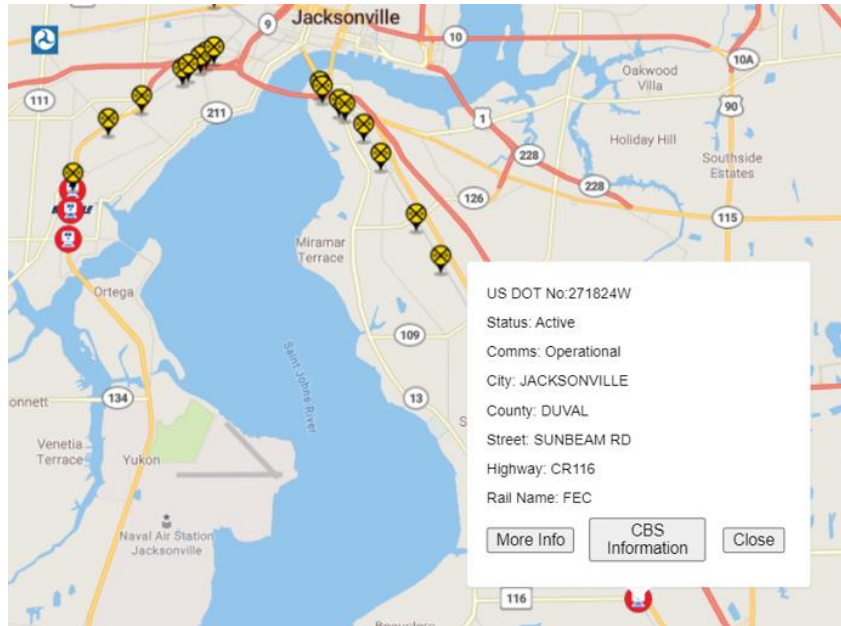
The yellow icons indicate HRIs in the area. When the user selects one of these icons a pop-up window with HRI information appears.



Source: Battelle

Figure 13. HRI Information

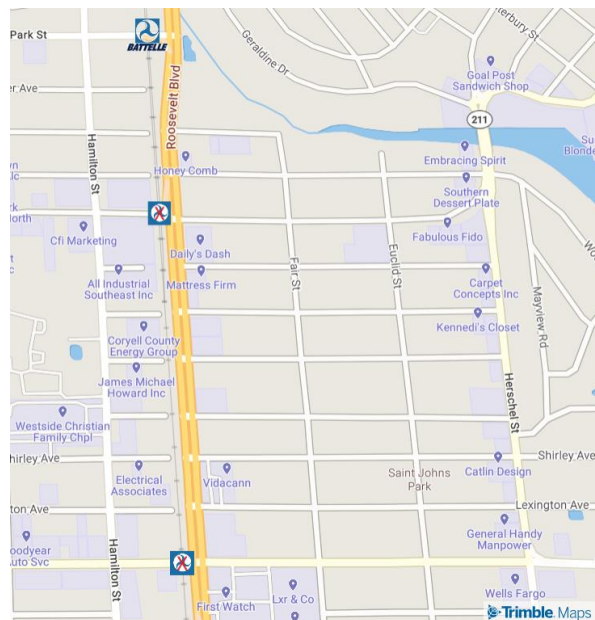
Blue icons indicate HRIs that are being monitored by the CBS. When the railroad preemption signal changed indicating the activation of the HRI warning devices, the icon changes and displays a red color.



Source: Battelle

Figure 14. CBS Equipped HRIs

Icons with an overlaying “X” indicate a loss of communications connectivity between the CBS and the HRI location.



Source: Battelle

Figure 15. Loss of Comms to Equipped HRIs

Web application Interface to the Cloud

The Web application is given a login identifier to be able to access the CBS API, which contains all the available information for any given HRI known to FRA, including those that are equipped with an RBS.

This location supports RESTful query options which enables access to data, tables and views. The following is an example of a RESTful query done to the CBS:

[https://rcvw-cbs-api.lemonmushroom-fb00896d.eastus.azurecontainerapps.io/api/Crossings?\\$select=HRI_ID,CrossingID,StateName,CountyName,Highway,Street&\\$filter=StateName%20eq%20%27FL%27%20and%20HRI_ID%20gt%200&uuid=demo](https://rcvw-cbs-api.lemonmushroom-fb00896d.eastus.azurecontainerapps.io/api/Crossings?$select=HRI_ID,CrossingID,StateName,CountyName,Highway,Street&$filter=StateName%20eq%20%27FL%27%20and%20HRI_ID%20gt%200&uuid=demo)

This query returns the following information in JSON format (Figure 16):

```
{
  "value": [
    {
      "HRI_ID": 27182,
      "CrossingID": "271824W",
      "StateName": "FL",
      "CountyName": "DUVAL",
      "Highway": "CR116",
      "Street": "SUNBEAM RD"
    },
    {
      "HRI_ID": 21221,
      "CrossingID": "621221S",
      "StateName": "FL",
      "CountyName": "DUVAL",
      "Highway": "NA",
      "Street": "ST JOHNS AVE"
    },
    {
      "HRI_ID": 21223,
      "CrossingID": "621223F",
      "StateName": "FL",
      "CountyName": "DUVAL",
      "Highway": "SR-128",
      "Street": "SAN JUAN AVE"
    },
    {
      "HRI_ID": 21275,
      "CrossingID": "621275X",
      "StateName": "FL",
      "CountyName": "DUVAL",
      "Highway": "US 95, SR111",
      "Street": "S EDGEWOOD AVE"
    }
  ]
}
```

Source: Battelle

Figure 16. Crossing Information Query Result

Another example of a subscribed user querying the current operational and preemption status for any particular HRI (once known) can be determined by periodically querying the API:

[https://rcvw-cbs-api.lemonmushroom-fb00896d.eastus.azurecontainerapps.io/api/Crossings?\\$select=HRI_ID,PreemptionStatus,RBSOperational&\\$filter=HRI_ID%20eq%2021275&uuid=demo](https://rcvw-cbs-api.lemonmushroom-fb00896d.eastus.azurecontainerapps.io/api/Crossings?$select=HRI_ID,PreemptionStatus,RBSOperational&$filter=HRI_ID%20eq%2021275&uuid=demo)

```

{
  "value": [
    {
      "HRI_ID": 21275,
      "PreemptionStatus": false,
      "RBSOperational": false
    }
  ]
}

```

Source: Battelle

Figure 17. HRI Status Query Result

However, for this demonstration the database queries were done directly without the use of the API. Additionally, the CBS is capable of pushing notifications of events such as operational state changes and pre-emption activation/deactivation at the crossings through the use of the Azure Service Bus to the Web Application:

[https://rcvw-cbs-api.lemonmushroom-fb00896d.eastus.azurecontainerapps.io/api/Crossings?\\$select=HRI_ID,Endpoint,Topic,Subscription&\\$filter=StateName%20eq%20%27FL%27%20and%20HRI_ID%20gt%200&uuid=demo](https://rcvw-cbs-api.lemonmushroom-fb00896d.eastus.azurecontainerapps.io/api/Crossings?$select=HRI_ID,Endpoint,Topic,Subscription&$filter=StateName%20eq%20%27FL%27%20and%20HRI_ID%20gt%200&uuid=demo)

The results of that push can be seen in Figure 18.

```

{
  "value": [
    {
      "HRI_ID": 27182,
      "Endpoint": "sb://rcvwcbssb.servicebus.windows.net/",
      "Topic": "cbs.events",
      "Subscription": "Sunbeam-Phillips"
    },
    {
      "HRI_ID": 21221,
      "Endpoint": "sb://rcvwcbssb.servicebus.windows.net/",
      "Topic": "cbs.events",
      "Subscription": "Roosevelt-StJohn"
    },
    {
      "HRI_ID": 21223,
      "Endpoint": "sb://rcvwcbssb.servicebus.windows.net/",
      "Topic": "cbs.events",
      "Subscription": "Roosevelt-SanJuan"
    },
    {
      "HRI_ID": 21275,
      "Endpoint": "sb://rcvwcbssb.servicebus.windows.net/",
      "Topic": "cbs.events",
      "Subscription": "Beaver-Edgewood"
    }
  ]
}

```

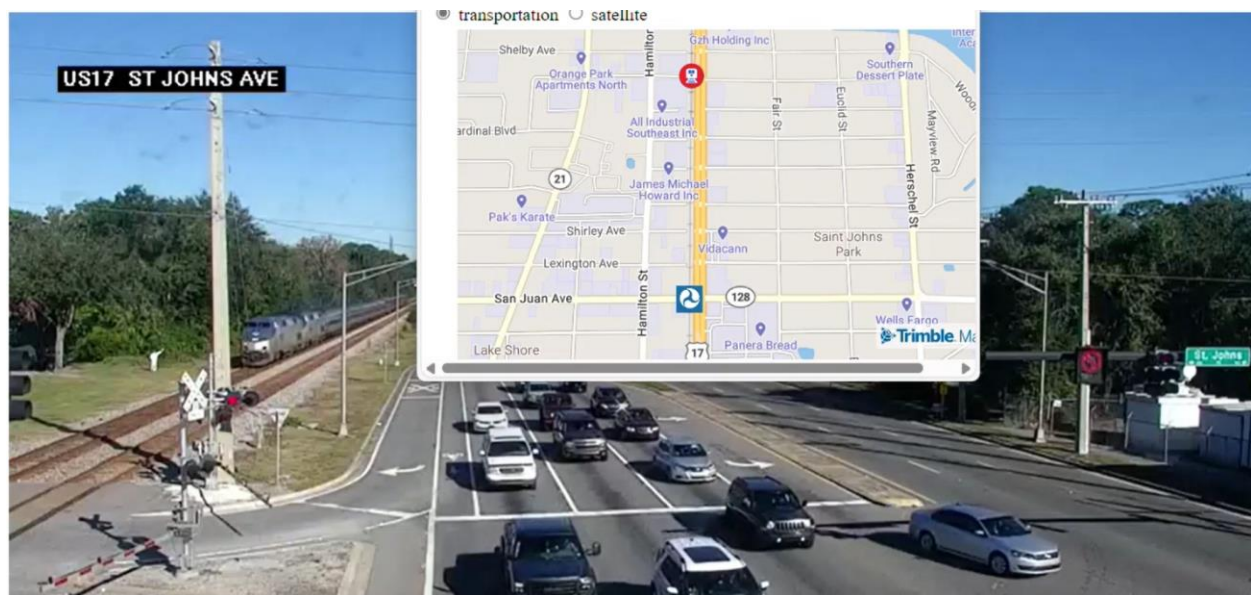
Source: Battelle

Figure 18. Results of Pushing Notifications of Events from the CBS

These are just a few examples of how CBS data may be retrieved. It is up to a client developer on how to connect to the service bus, and a different Read or Read/Write CBS subscription login will be needed to access the topics.

Traffic Camera Video Display

To be able to verify that train presence at the monitored locations were being correctly registered by the CBS, FDOT provided access to a traffic camera located at the St. Johns location. This tool allowed the Battelle team to visually verify when the web service indicated an activation of the warning devices. Figure 19 presents a train arriving to the St. John's HRI. The active warning devices are activated, and the web service has indicated (by changing the icon to a red color and adding a shape of an incoming train) that an incoming train has been detected.



Source: Battelle

Figure 19. Train approaching St. Johns location.

For each of the four monitored sites, FDOT provided daily records of train presence at the HRIs to compare with the CBS logs. The data collected by the CBS for each of these four locations was used to produce an analysis of the latencies that the system experiences. See the **Error! Not a valid bookmark self-reference.** section for more details.

During the time the CBS was active in the city of Jacksonville it collected the following information

- Date and time of warning device activation
- Date and time of warning device deactivation
- Communications status
- All Communication logs between RBS and CBS

The Communication logs between the RBS and the CBS are collected on a 24/7 basis. These logs provide information regarding each HRI location such as:

- Time a status message was created in the RBS
- Time a status message was received at the CBS
- Warning device activation/deactivation

This data was used to produce an analysis of the latency encompassing the time from the detection in the loss of the pre-emption signal through the time it is registered and processed in the CBS.

This section details the process undertaken to analyze the performance of message transmission from the RBS to the CBS.

Data Collection

The analysis began with measuring the time intervals associated with message transmission. Two key metrics were utilized:

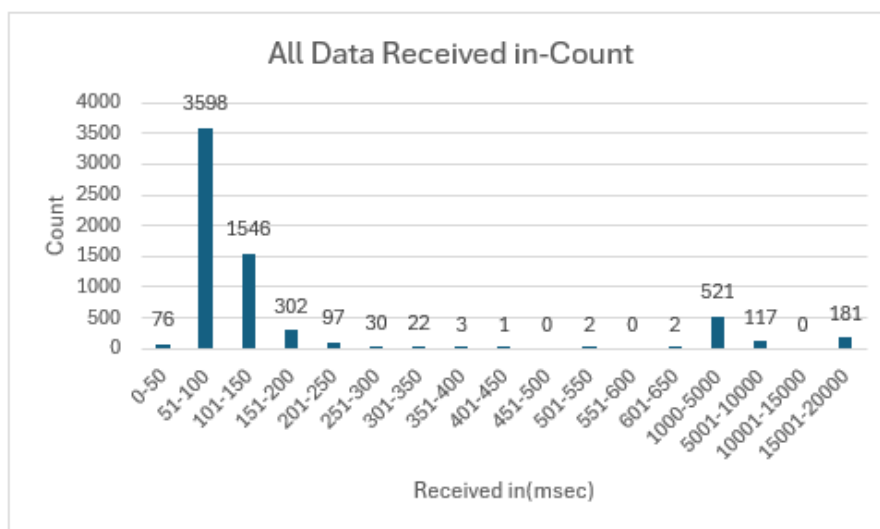
- Received Time = (Time of Message received at CBS) – (Time of Message created at RBS)
- Stored Time = (Time of Message stored at CBS) – (Time of message created at RBS)

Given that direct measurement of transit time from RBS to CBS was not feasible, the message creation time was used as a conservative estimate. This estimate was then compared against the CBS requirement of 0.5 seconds (500 milliseconds) for message receipt and 5 seconds (5000 milliseconds) for the entire process, including CBS ingest performance. Efforts were made to ensure that any discrepancies in clock synchronization between CBS and RBS were negligible, within a range of 1-2 milliseconds. The analysis used a total of 8917 random records from all four sites. This data was extracted from 12 randomly selected days. This approach ensured a statistically valid sample size. This sampling method was designed to enhance the reliability of the statistical analysis, ensuring that data from all relevant sites were adequately represented.

Data Analysis

The collected data were compiled into a spreadsheet for detailed analysis. Critical performance metrics were calculated, including minimum, maximum, average, and percentile measurements (50th, 75th, and 95th percentiles) for the RBS to CBS communications. The 95th percentile value was specifically used to validate compliance with the performance requirements. This was done for each HRI as well as the combined data.

	(msec)
Average:	877.3457
Max:	17230
Min:	40
50th percentile:	87
70th percentile:	130
95th percentile:	2330



Source: Battelle

Figure 20. All Data, Received in count

(msec)

Average: 1006.057

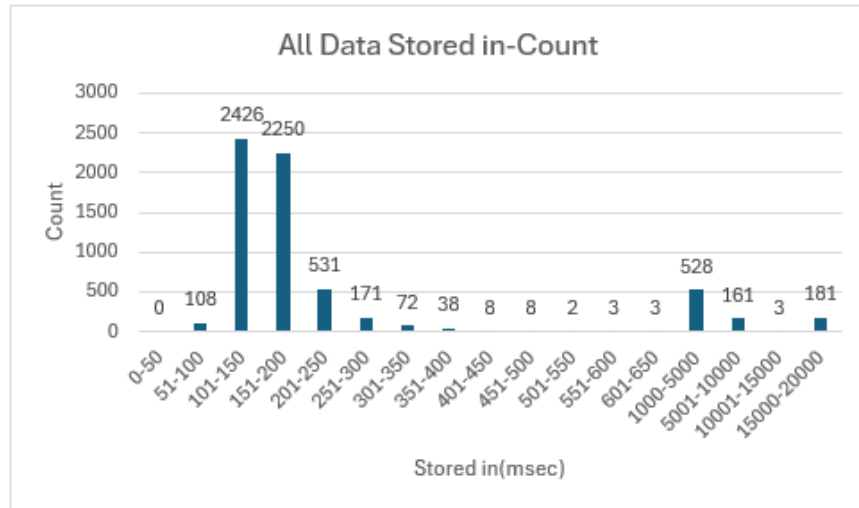
Max: 17503

Min: 93

50th percentile: 167

70th percentile: 194

95th percentile: 6028.5



Source: Battelle

Figure 21. All Data, Stored in count

Source: Battelle

Figure 22 to Figure 29 show the data analyzed per location.

(msec)

Average: 89.4846765

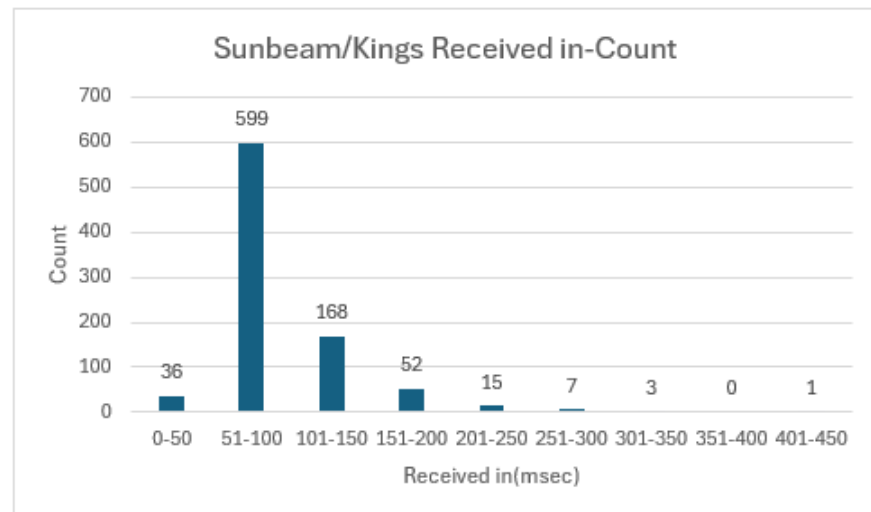
Max: 443

Min: 44

50th percentile: 73

70th percentile: 90

95th percentile: 170



Source: Battelle

Figure 22. Sunbeam/Kings Data, Received in count

(msec)

Average: 154.3473326

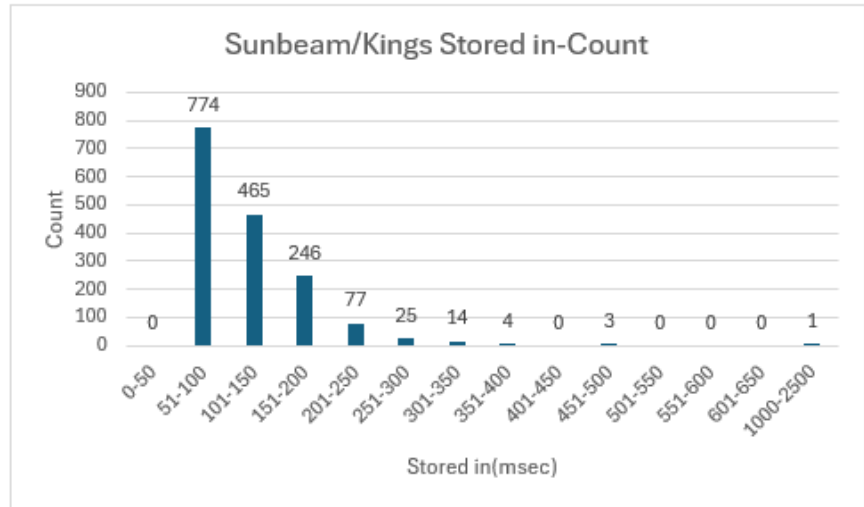
Max: 1200

Min: 93

50th percentile: 140

70th percentile: 167

95th percentile: 253



Source: Battelle

Figure 23. Sunbeam/Kings Data, Stored in count

(msec)

Average: 77.48258

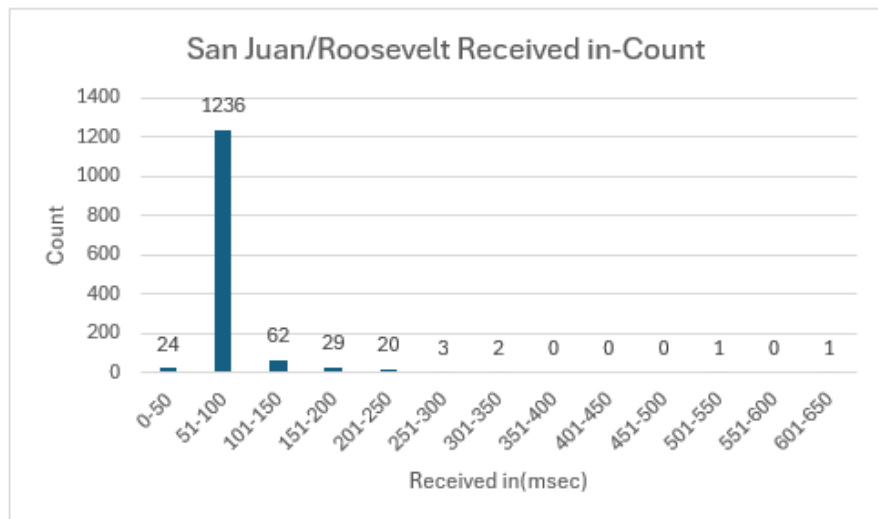
Max: 650

Min: 40

50th percentile: 70

70th percentile: 77

95th percentile: 133



Source: Battelle

Figure 24. San Juan/Roosevelt Data, Received in count

(msec)

Average: 257.7322

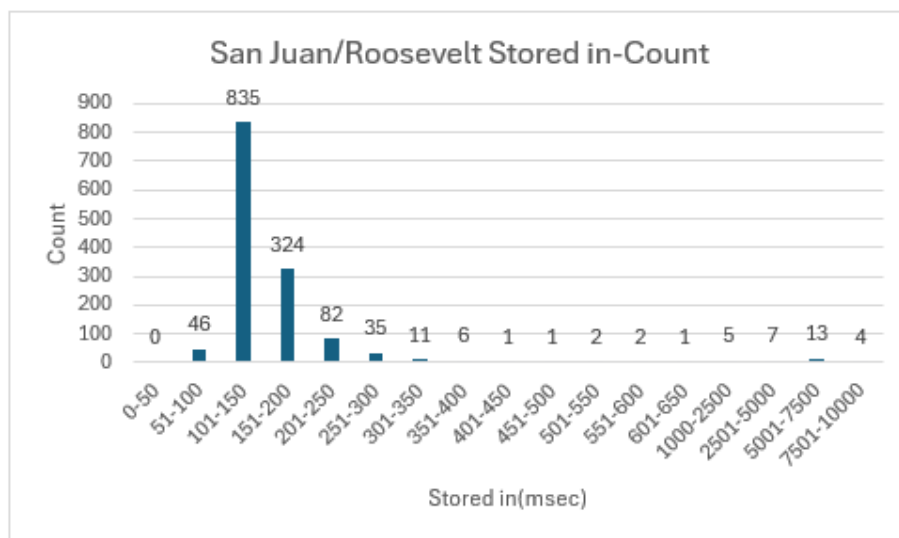
Max: 9563

Min: 93

50th percentile: 136

70th percentile: 160

95th percentile: 284



Source: Battelle

Figure 25. San Juan/Roosevelt Data, Stored in count

(msec)

Average: 2787.03673

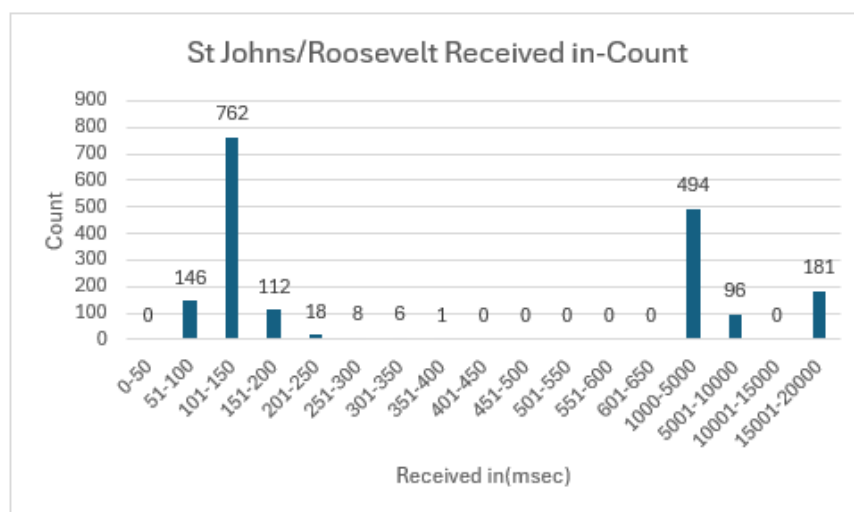
Max: 17230

Min: 70

50th percentile: 153

70th percentile: 2017

95th percentile: 17060



Source: Battelle

Figure 26. St. Johns/Roosevelt Data, Received in count

(msec)

Average: 2921.81689

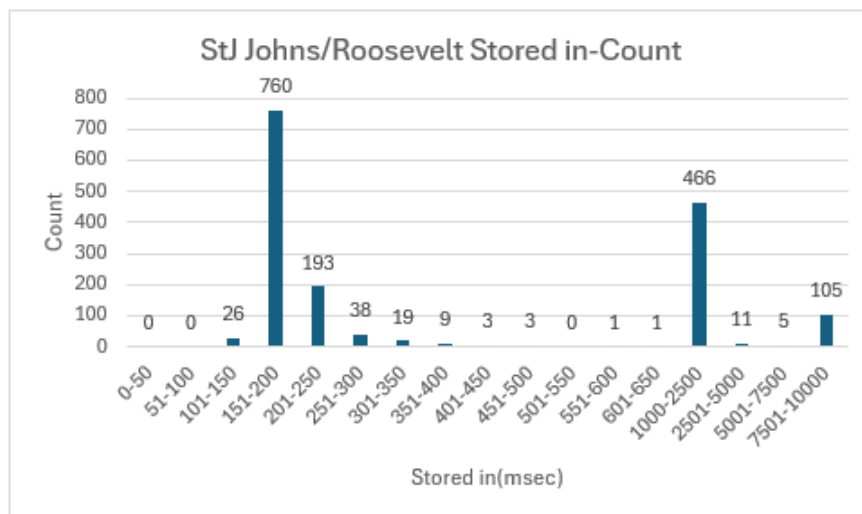
Max: 17503

Min: 116

50th percentile: 220

70th percentile: 2100

95th percentile: 17164



Source: Battelle

Figure 27. St. Johns/Roosevelt Data, Stored in count

(msec)

Average: 179.3897

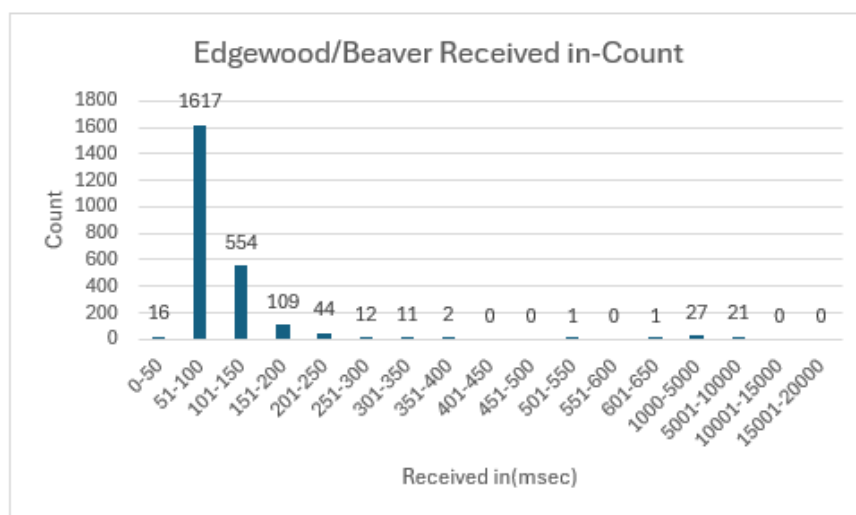
Max: 7383

Min: 40

50th percentile: 80

70th percentile: 107

95th percentile: 200.6



Source: Battelle

Figure 28. Edgewood/Beaver Data, Received in count

(msec)

Average: 297.41

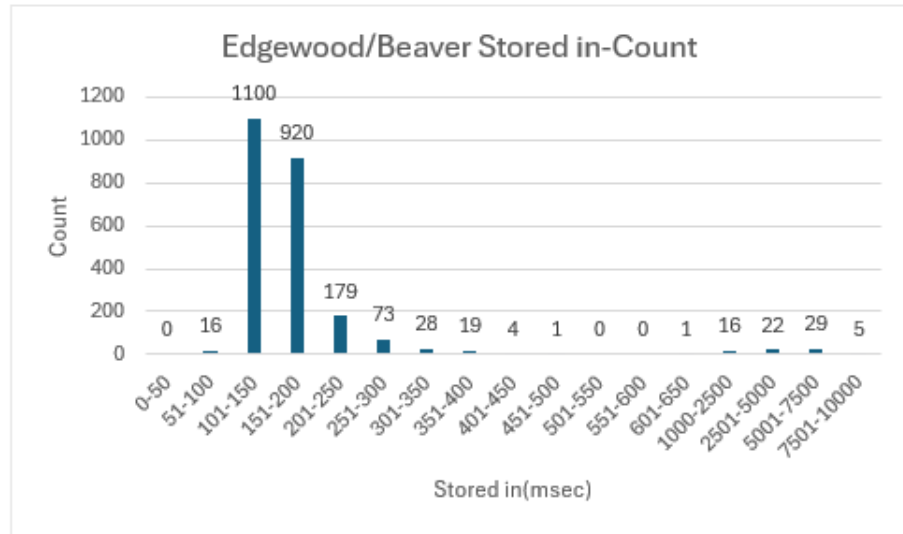
Max: 8697

Min: 93

50th percentile: 154

70th percentile: 174

95th percentile: 310



Source: Battelle

Figure 29. Edgewood/Beaver, Stored in count

Source: Battelle

Figure 20 and Source: Battelle

Figure 21 show that data latency, when analyzed for the total set, fails to meet the corresponding latency requirements. These latencies are the result of internet connection issues at St. John's/Roosevelt crossings. The three remaining crossings show no latency issues (Figure 26 and Figure 27). The Battelle team decided to remove the wired internet connection at this location and replace it with a portable cellphone router. The change was made on October 17, 2024, and the result was a significant increase in communication speed. Figure 30 and Figure 31 show the St. John's/Roosevelt data before and after October 17.

(msec)

Average: 5235.905

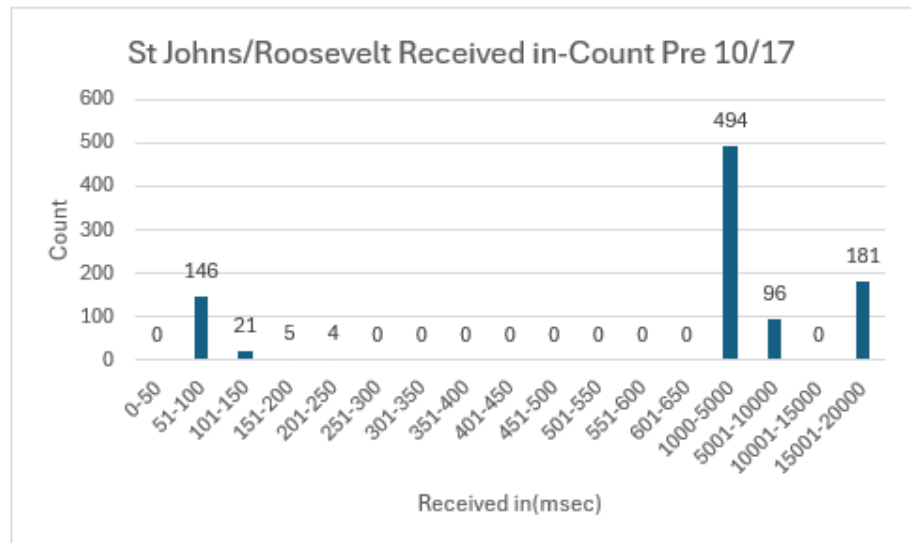
Max: 17230

Min: 70

50th percentile: 2047

70th percentile: 2371.4

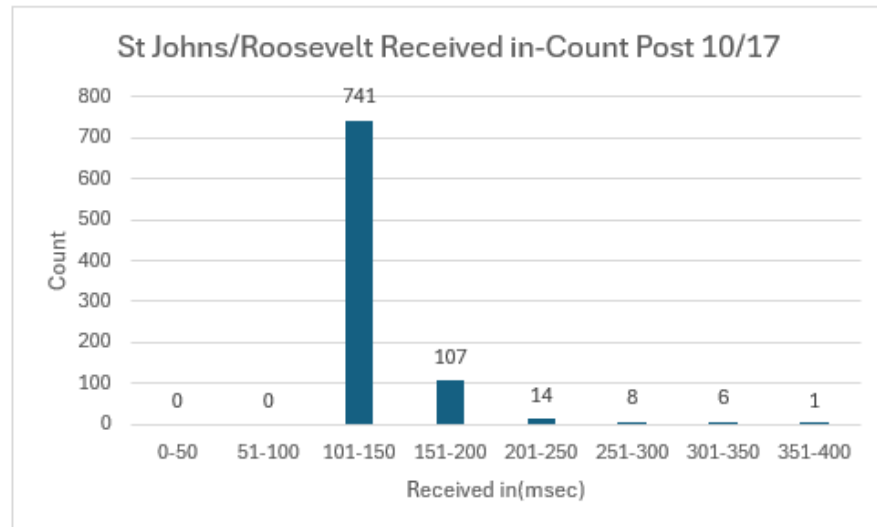
95th percentile: 17064



Source: Battelle

Figure 30. St. John's/Roosevelt Data Before 10/17, Received in count

	(msec)
Average:	142.7058
Max:	367
Min:	107
50th percentile	137
70th percentile	146
95th percentile	173



Source: Battelle

Figure 31. St. John's/Roosevelt Data After 10/17, Received in count

Overall, all four crossings were well within the required thresholds for both receiving and storing messages from the RBS. As seen in the above graphs, after placing the modem in the St. John's/Roosevelt location, its performance was on par with the other three crossings.

An analysis was also done to confirm the RBS activations matched those that were provided to Battelle by FDOT. During an analysis of 10/18/2024 through 11/14/2024, the CBS correctly matched 1746/1792 activations/deactivations (97.4%) during days when the systems were actively tracking crossing activity. A probable reason for the missed activations/deactivations is hypothesized to be the result of a periodic disconnection of the RBS to the CBS. See Challenges and Solutions section for information.

Challenges and Solutions

There were a few challenges the Battelle team faced during this demonstration of the RCVW CBS.

Environmental Issues

During the demonstration period, the CBS experienced periodic communications issues with the equipment installed at the four traffic signal cabinets (see Periodic Disconnection of the RBS to the CBS). A proposed solution to mitigate this problem in one location was to install a mobile network hotspot to allow a wireless internet connection with the RBS computer.

The Battelle team installed a small portable cellphone router (Verizon Jetpack® MiFi® 8800L) to achieve this. Communication issues were still experienced after the implementation of the cellphone router. After an analysis of the situation, it was discovered that extreme temperatures inside the traffic signal cabinet in Florida were affecting the functionality of the mobile hotspot.

It was suggested by FDOT personnel to replace the device with a rugged Long Term Evolution (LTE) router that could withstand the temperatures inside the cabinet. The Battelle team replaced the device with a BR1 Mini LTE router with an operating temperature rating of temperature -40°C to 65°C. After this, there were no further temperature related connection issues.

Periodic Disconnection of the RBS to the CBS

Following periods of network disruption, as described above, the RBS should automatically reconnect to the CBS to continue delivery of the messages. However, this failed to occur during the period of demonstration. The issue was due to a low-level technical constraint in the third-party software (Apache Qpid Proton) used to connect and deliver messages to CBS via the Advanced Messaging Queuing Protocol (AMQP). Each message delivery is associated with an AMQP *sender*, but the RBS code was originally written so that each sender utilizes a single *connection* object. While a connection for Apache Qpid Proton contains automatic reconnection capabilities, those capabilities are not coded into RCVW correctly. Therefore, when the connection fails due to a disruption, no future senders are created. To remedy this, a workaround to periodically restart the HRI plugin on the RBS every 30 minutes (at :15 and :45 past the hour) was implemented in order to ensure new connections were established. The restarts were relatively fast and were not considered to affect data analysis during normal operating periods. However, post-data collection analysis was needed to filter-out clear indicators that a restart was occurring.

Late in the demonstration cycle, Batelle inserted code to build a sender with its own connection object, making it possible for automatic reconnection to occur. However, no time was permitted to validate this change on the sites in Jacksonville.

Time Synchronization

The current CBS configuration and communications interface does not allow for direct time synchronization¹ with the rest of the traffic management devices. This resulted in the inability to perform a complete analysis of the performance of the system in comparison to the tools currently used.

To achieve this, the CBS needs to be fully integrated into the organization's network and allowed to be managed from within. However, even that the CBS was not considered as part of the network architecture, the system was proven to be reliable and accurate.

Signal Preemption Interface

Due to concerns that the RCVW would disrupt operation at the Traffic Signal Controller, a direct connection to the railroad preemption signal was not possible. Instead, the city of Jacksonville proposed the use of a load switch which activates a Blank out sign when pre-emption changes occurred. While this was not the way the system was designed to operate, it prove to be an effective way to indirectly receive the railroad signal preemption.

There is always a safety concern when a system interacts with Railroad and Highway devices. During this demonstration it was proven that there are methods that address and reduces these safety concerns which allow the RCVW to perform its functions without directly interfacing with the HRI controller or Traffic Signal Controller.

¹ For example, the RBS and VBS systems use the Linux *chrony* tool (<https://chrony-project.org/>)

Advanced Railroad Preemption vs Simultaneous Railroad Preemption

There are two types of railroad preemption, Advanced Railroad Preemption and Simultaneous Railroad Preemption. Advanced Railroad Preemption involves activating vehicle traffic signals at a set time before the railroad crossing signals activate. The goal is to clear any vehicle traffic that may be queued at the crossing ahead of the train's arrival, ensuring that no vehicles are trapped on the tracks. Simultaneous Railroad Preemption activates the traffic signals at the same time as the railroad crossing signals are activated. Traffic signals immediately go into a preemption sequence to stop all road traffic as soon as the train is detected, allowing for the barriers to come down and the train to pass safely. The RCVW CV application was designed to use simultaneous railroad preemption in order to generate warnings and alerts to the vehicle driver at the same time the warning devices had activated. In the case of this demonstration, the RCVW system interfaced to the preemption signal used by the traffic controller at each location which turned out to be advanced preemption. The Battelle team was not aware of this until videos of system activation were reviewed. Using the advanced preemption resulted in the system reporting activations of the HRI warning devices seconds before their actual activation.

It can be argued that for the case of the CBS, this is a desired feature since it would report to the user in advance of the warning devices activation, but a more in-depth analysis needs to be performed to fully understand the benefits or disadvantages of these two types of preemptions when used with the RCVW system.

Conclusions

This demonstration event provided enough data and information confirming that the RCVW:

- through the CBS is capable of reporting preemption signal changes to subscribed users over the internet.
- is capable of interfacing in multiple ways with traditional traffic management devices in the field to obtain HRI activation/deactivation status.
- through the CBS, is capable of receiving and processing incoming RBS messages and storing the related information in a dedicated database.
- through the CBS is capable of storing static HRI data in a dedicated database.
- provides the capability for a user to query the historical data stored in the CBS.
- RBS is capable of receive train preemption data via diverse sources. In the case of this demonstration, the RBS received this information from the Traffic Signal Controllers which in turn received this data from the railroad crossing controllers.
- is capable of broadcasting timely crossing data beyond the CV technology.
- does not require the installation of additional train sensing equipment.
- seamlessly interface with existing DOT and municipal traffic equipment with no disruption to the traffic management activities.
- the processing time of detecting and reporting a train presence in an HRI complies with the CBS requirements.

It was shown in this demonstration that the deployment of the CBS can be relatively simple, with no disruption to the traffic management activities. This demonstration only highlighted the use of the CBS as a tool for a TMC. However, the fact that any type of user (with the proper credentials) can have access to real time HRI status over the internet allowed for multiple use cases such as

- Vehicle Navigation Apps
- Freight Dispatchers
- In-vehicle digital alerts
- Emergency Vehicle routing

Next Steps

To fully make use of the capabilities of the CBS functionality, the results of this research should be disseminated to the public and related industries. A way to do this is to participate in industry related conferences and meetings where the results of this demonstration can be presented and highlighted. A few examples of this are:

- Transportation Research Board Annual Meeting
- American Railway Engineering and Maintenance-of-Way Association (AREMA) Annual Conference and Expo
- Governors Highway Safety Association Annual Meeting
- ASCE International Conference on Transportation

Reaching out to potential users of the CBS such as vehicle navigation system developers, freight dispatch software companies and State DOTs could help disseminate the use of its functionality. A few examples are:

- Waze
- HASS Alert
- Google Maps
- Omnitrac
- Trimble
- Axon

Additional research and analysis on the use of advanced preemption vs simultaneous preemption needs to be conducted to fully understand the cases where one or the other should be used and create methods that allow the deployment of RCVW regardless of the preemption type being used.

Acronyms

ACRONYMS

ASCE

API

AMQP

AREMA

CBS

CCTV

CV

FDOT

FRA

GUI

HRI

ID

LTE

MAP

WiFi

MUTCD

NHTS

RBS

RCVW

SPaT

TMC

VAC

VDC

EXPLANATION

American Society of Civil Engineers

Application Programming Interface

Advanced Messaging Queuing Protocol

American Railway Engineering and Maintenance-of-Way Association

Cloud-based Subsystem

Closed-Circuit Television

Connected Vehicle

Florida Department of Transportation

Federal Railroad Administration

Graphical User Interface

Highway-Rail Intersection

Identification Number

Long Term Evolution

HRI geometry message

Portable Wireless Router Over Cellular Network

Manual on Uniform Traffic Control Devices

National Household Travel Survey

Roadside-Based Subsystem

Rail Crossing Violation Warning

Signal Phase and Timing

Traffic Management Center

Volts Alternate Current

Volts Direct Current